

MICROMOLD · DESIGN DECISIONS REGISTER

MicroMold design decisions register

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DOCUMENT CONTROL

Document ID	MMD-DEC-001	Version	0.3
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/micromold		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Value engineering updated for the torque-limiting socket and plunger rest (press USD 590, USD 70 over the target)
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 7 on 2026-10-02 (MMD-DDR-003 accepted with P1 confirmed against the bought press, lift-out loading with a plunger rest, bench hole, torque-limiting socket, Precious Plastic workspace near Irving as first candidate, first product mold, appearance model); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

MicroMold design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The arbor press: a 28 mm ram 255 mm or longer with 150 mm of engaged travel; a head that can be cut from its frame with a flat back face and enough metal for four M10 holes 20 deep; the head's mass (8 kg assumed)	Sets the stroke, the head fixing and R9's mass margin (1.0 kg)	MMD-DDR-003, P1; MMD-CAL-001 [K1]
2	The press's lever hub: its outside diameter and bore, to take the welded square drive	The ratchet adapter is made from it	MMD-DDR-003, P13
3	The load cell is about 56 mm across, has a top M12 stud and a base flange with three M5 through holes (46 mm circle assumed)	The spacer's tapped holes are drilled to match it	MMD-DDR-003, P6
4	The handwheel's hub takes a Tr20 x 4 nut (about 36 mm across)	The handwheel nut is made from it	MMD-DDR-003, P7
5	The mold cooling fan's corner holes are on a 105 mm square	The fan bracket's holes	MMD-DDR-003, P10
6	The nozzle heater band fits a 24 mm body and is no more than 18 mm wide	The nozzle body length	MMD-DDR-003, P4
7	The bench top's thickness and the clear space under it (60 mm or more below the hole)	The lift screw drops through it	MMD-DDR-003, P7

■ Value engineering

Value-engineering target: USD 520 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 590 (USD 70 over the target) for the press; USD 684 with the test mold, which is tooling outside the target (USD 94, R8). Main cost drivers and savings worth trying:

- The largest lines are the arbor press head with column tube and ratchet (USD 110), the control box (USD 55), the fume hood and duct fan (USD 48), the barrel (USD 45), the load cell and display (USD 35) and the lift table, screw and handwheel (USD 30).
- Making the design constructable added USD 37: the construction parts of line 19 (USD 30: head mounting plate and screws, plunger coupling and ball-lock pin, ratchet adapter, hood arm and rivnuts), the bracket's back plate

and cap screws (USD 4) and the shield's hinges and latch (USD 3). The mold rose USD 4 for screws, inserts and dowels.

- The decisions of 2026-10-02 added USD 46: the torque-limiting socket (line 21, USD 40, an estimate at the midpoint of a USD 30 to 50 retail band) and the plunger rest (line 20, USD 6, an estimate by parts).
- Savings worth trying: a used arbor press (often under half the new price); counting the duct fan and duct (USD 48) as workshop extraction shared with other machines, as the molds are counted as tooling, which would bring the press to USD 542 (that removes cost from scope rather than saving it, so it does not compare like with like with the target); buying the two PID controllers and relays as a pre-wired dual-zone controller; and a ball-lock pin replaced by a plain clevis pin and R-clip (about USD 4 less).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: rack-and-pinion drive from a 1 t arbor press on a taller column, vertical barrel with the mold below, plunger injection, two PID zones, bolted two-plate molds on a screw lift table, 10 kN load cell, test plaque first, molds as tooling, side hood with duct fan, a partner that already shreds HDPE or PP	Amish: "i accept all your recommendations, go with them across all repos."	MMD-DDR-001, MMD-DDR-002
2026-09-25	N1 to N5: budget USD 500 for the press, 40 kg mass target, two 300 W barrel bands, a mold cooling fan, injection-grade flake only	Amish, same instruction	MMD-DDR-002
2026-09-26	Budget USD 520 to cover the priced BOM (O3, option (a))	Amish: "i approve all the budget items."	MMD-DDR-002 v0.2
2026-09-30	Make the design physically buildable as the build plan is drawn; changes recorded as draft for review	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	MMD-DDR-003 (its changes were accepted on 2026-10-02, below)
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	This register
2026-10-01	<code>budget_usd</code> is a hypothetical value-engineering target, not a spending limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register; MMD-REQ-001 R14
2026-10-02	Design for construction accepted: the changes P1 to P15, with one condition: P1 (cutting the head from the arbor press casting and tapping four M10 holes in its back) is confirmed against the bought press before it is cut	Amish: "i approve your recommendations for all 555 open decisions."	MMD-DDR-003, Table 1
2026-10-02	Loading: option (a), pull the ball-lock pin and lift the plunger out to load, then refit it to tamp and inject; a simple plunger rest is added on the column so the hot plunger is never laid on the bench	Amish: "i approve your recommendations for all 555 open decisions."	MMD-DDR-003, A1
2026-10-02	Lift screw: option (a), a 25 mm hole through the bench for the lift screw is accepted	Amish: "i approve your recommendations for all 555 open decisions."	MMD-DDR-003, A2
2026-10-02	Overload protection: option (b), a torque-limiting socket on the ratchet adapter, set so the ram cannot exceed the press rating, and checked against the load cell at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	MMD-DDR-003, A3; docs/REVIEW.md, TRL 3 safety concerns
2026-10-02	First co-design partner: a community plastic recycling workspace that already shreds HDPE or		MMD-DDR-001, O1

Date	Decision	Decided by	Record
	PP; first candidate to approach a Precious Plastic workspace near Irving, Texas	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-02	First product mold after the test plaque: a flat, single-cavity part of 25 g or less that fits the 120 x 90 mm mold plates, such as a coaster, tile or cable clip, chosen with the partner from what it can sell	Amish: "i approve your recommendations for all 555 open decisions."	MMD-DDR-001, 02
2026-10-02	Appearance model: accept differences 1, 3, 5 and 6; update 2 and 4 to the constructable design (no guide rods; cap screws into inserts, no nuts), and add the screw lift, flange and fan bracket at the same time	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26, items 1 to 6